

LowMachReact-Hex Technical Overview

Architecture, Numerics, Thermodynamics, and Performance — v0.5.0

LowMachReact-Hex Development Notes

May 2026

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Current Solver Interpretation

Core statement

LowMachReact-Hex is currently a constant-density low-Mach / incompressible finite-volume projection solver on hexahedral meshes.

- Optional passive species mass-fraction transport.
- Optional passive sensible-enthalpy transport.
- Optional Cantera thermodynamics and transport properties.
- MPI-aware diagnostics, VTU/PVTU output, and profiling.

$$\rho_f = \rho_{\text{params}}$$

Important convention

Cantera thermodynamic density ρ_{thermo} is diagnostic / future-use only and is not used by the pressure projection.

Currently enabled

- Constant-density projection.
- Passive species transport.
- Passive sensible enthalpy.
- Heat conduction via ∇T .
- Cantera h , T , c_p , λ , μ , D_k .
- Diagnostic q_{rad} storage.

Not yet implemented

- Variable-density low-Mach coupling.
- Reactions.
- Reaction heat release.
- Species-diffusion enthalpy correction.
- External radiation physics.
- Radiation spectral decomposition.

Module Organization

Module	Responsibility
<code>mod_input</code>	Namelist parsing and validation
<code>mod_mesh_types/io</code>	Mesh structures and native mesh import
<code>mod_bc</code>	Boundary-condition mapping and face states
<code>mod_fields</code>	Velocity, pressure, divergence, face fluxes
<code>mod_flow_projection</code>	Momentum prediction and pressure projection
<code>mod_species</code>	Passive species transport
<code>mod_energy</code>	Sensible enthalpy and Cantera thermo sync
<code>mod_transport_properties</code>	Constant / Cantera transport properties
<code>mod_mpi_flow</code>	Replicated-mesh MPI utilities
<code>mod_output/profiler</code>	VTU/CSV output and timing reports

Mesh pipeline

Gmsh mesh $\rightarrow$ meshio conversion $\rightarrow$ native files $\rightarrow$ Fortran mesh_t.

- Every MPI rank stores the full mesh.
- Flow work is split by contiguous owned cell ranges.
- Owned cell updates are synchronized with exchanges, gathers, and collective reductions.
- The replicated mesh keeps the future radiation decomposition simple, because every rank can see full geometry.

$$\text{rank } r : \quad c \in [\text{flow\%first_cell}, \text{flow\%last_cell}]$$

Incompressible Flow Model

$$\begin{aligned}\nabla \cdot \mathbf{u} &= 0, \\ \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) &= -\frac{1}{\rho_f} \nabla p + \nabla \cdot (\nu \nabla \mathbf{u}) + \mathbf{f}.\end{aligned}$$

- ρ_f is always the configured flow/projection density.
- Cantera may supply μ , but variable-viscosity flow is optional.
- If variable viscosity is enabled:

$$\nu = \frac{\mu}{\rho_f}.$$

Passive Species and Sensible Enthalpy

Species mass fraction

$$\frac{\partial Y_k}{\partial t} + \nabla \cdot (\mathbf{u} Y_k) = \nabla \cdot (D_k \nabla Y_k), \quad \dot{\omega}_k = 0$$

Sensible enthalpy

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u} h) = \frac{1}{\rho_f} \nabla \cdot (\lambda \nabla T) + \frac{q_{\text{rad}}}{\rho_f}$$

Energy convention

h is transported. T is recovered from h , Y , and p_0 . Conduction uses ∇T , not ∇h .

Finite-Volume Update

For cell c , face f , volume V_c , and outward local flux $F_{f,c}$:

$$\psi_f^{\text{up}} = \begin{cases} \psi_c, & F_{f,c} \geq 0, \\ \psi_{\text{other}}, & F_{f,c} < 0. \end{cases}$$

- Stored face flux is owner-to-neighbor.
- During cell updates it is reoriented outward from the current cell.
- Diffusive face gradients use normal cell-to-cell or cell-to-boundary distances.
- Species are bounded and renormalized after explicit update.

Projection Algorithm

1 Predict intermediate velocity

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} = \mathcal{R}^{n,n-1} - \frac{1}{\rho_f} \nabla p^n.$$

2 Solve pressure correction

$$\nabla^2 \phi = \frac{\rho_f}{\Delta t} \nabla \cdot \mathbf{u}^*.$$

3 Correct velocity and pressure

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{\Delta t}{\rho_f} \nabla \phi, \quad p^{n+1} = p^n + \phi.$$

Sensible Enthalpy Convention

Cantera absolute enthalpy includes formation contributions. The solver stores sensible enthalpy relative to T_{ref} :

$$h(T, Y, p_0) = h_{\text{abs}}^{\text{Cantera}}(T, Y, p_0) - h_{\text{abs}}^{\text{Cantera}}(T_{\text{ref}}, Y, p_0).$$

Temperature recovery reconstructs an absolute enthalpy target:

$$h_{\text{target,abs}} = h + h_{\text{abs}}^{\text{Cantera}}(T_{\text{ref}}, Y, p_0).$$

Then Cantera HPY inversion recovers T .

Option A: Preserve Transported Enthalpy

After species transport

Species may change from Y^n to Y^{n+1} . The solver preserves transported h and recovers a new temperature:

$$T^* = T(h, Y^{n+1}, p_0).$$

Do not do this

Do not recompute enthalpy from old temperature and new composition:

$$h^* = h(T^n, Y^{n+1}, p_0).$$

This avoids adding or removing sensible enthalpy without an explicit energy flux or source.

Thermo Sync and Density Split

Preferred energy-path sync:

$$(T, c_p, \lambda, \rho_{\text{thermo}}) = \text{sync}(h, Y, p_0).$$

- Sync must preserve transported h .
- Pre-flux sync is needed only when transported species may have changed Y .
- Species-disabled Cantera thermo can skip redundant pre-flux sync.
- Post-flux sync refreshes thermo state after h transport.
- `thermo_update_interval` is reserved and must remain 1.

$$\rho_f = \text{params}\%rho, \quad \rho_{\text{thermo}} = \text{diagnostic only.}$$

Main Timestep Ordering

- 1 Update CFL or timestep diagnostics when needed.
- 2 Refresh transport properties on `transport_update_interval`.
- 3 Advance momentum and pressure projection.
- 4 Advance passive species, if enabled.
- 5 Advance passive sensible enthalpy, if enabled.
- 6 Recover or synchronize T , c_p , λ , and ρ_{thermo} .
- 7 Write diagnostics and VTU/PVTU output on output steps.

Transport update

Cantera may update μ and D_k , but flow density remains $\rho_f = \text{params}\%rho$.

Boundary Conditions

Boundary patches are named in the mesh and configured through `case.nml`.

- Legacy patch type can act as fallback.
- Field-specific overrides exist for velocity, pressure, species, and temperature/enthalpy.
- Periodic connectivity can be loaded from native mesh data.

Fixed-temperature Cantera boundary

With species and Cantera thermo enabled:

$$h_b = h(T_b, Y_b, p_0).$$

Boundary composition Y_b comes from the species boundary condition.

Diagnostics and Profiling

- Outputs include VTU/PVTU visualization and CSV diagnostics.
- Energy output can include T , h , q_{rad} , c_p , λ , and ρ_{thermo} .
- Species output includes Y_k , $\sum_k Y_k$, and optionally D_k .
- Profiling reports inclusive wall-clock time.
- Flat profiler rows are not additive when nested timers are enabled.

Common timers:

`Projection_Step`, `Species_Transport`, `Energy_Transport`.

Build and Run

```
# Create once.
conda env create -f environment.yml

# Activate each session.
conda activate react_env

# Load OpenMPI if required on your system.
module load openmpi

# Build.
make BUILD=release

# Run.
mpirun -np 4 ./lowmach_react_hex cases/channel_flow/case.nml
```

- Use the top-level `./lowmach_react_hex` launcher.
- The launcher prioritizes Conda runtime libraries for Cantera.

Validation Ladder

- 1 Debug build and startup smoke test.
- 2 Single-rank incompressible flow case.
- 3 Multi-rank repeat of the same case.
- 4 Passive species with constant diffusivity.
- 5 Constant-property energy transport.
- 6 Cantera thermo without transported species.
- 7 Species-coupled Cantera thermo.
- 8 Profiling comparison for performance changes.

Keep for comparison

Case file, rank count, build mode, terminal output, CSV diagnostics, and representative visualization output.

Performance: Granular Sub-Module Profile

Sub-Module Breakdown (83.6 s total wall time)

- **Cantera PreSync (26.1%)**: Per-cell HPY inversion before flux computation.
- **Cantera PostSync (22.3%)**: Full property refresh after h update.
- **PCG MPI (21.1%)**: Global dot products $\rightarrow$ latency-bound.
- **Matvec (6.9%)**: Matrix-free Laplacian is efficient.
- **Energy Comm (6.9%)**: Sync of T, h, c_p, λ across ranks.

Sub-Region	%
Cantera PreSync	26.1
Cantera PostSync	22.3
PCG MPI Comm	21.1
Matvec	6.9
Energy Comm	6.9
Predictor	2.5
Species	1.8

Performance: Full Profiling Tree

Region Calls Avg(s) Avg%

```
-----  
Total_Simulation 1 83.564 100.00  
  Projection_Step 16000 33.994 40.68  
    Projection_PCG 16000 28.053 33.57  
      MPI_Communication 6629753 17.609 21.07  
      Pressure_Matvec 2199251 5.786 6.92  
    Projection_Predict_Flux 16000 2.105 2.52  
    Projection_Momentum_RHS 16000 1.732 2.07  
  Energy_Transport 16000 47.180 56.46  
    Energy_Cantera_PreSync 16000 21.847 26.14  
    Energy_Cantera_PostSync 16000 18.591 22.25  
    Energy_PreFlux_Exchange 16000 3.794 4.54  
      MPI_Communication 32000 3.773 4.52  
    Energy_Final_Exchange 16000 2.029 2.43  
      MPI_Communication 48000 2.002 2.40  
  Species_Transport 16000 1.531 1.83  
    MPI_Communication 16000 0.209 0.25
```

Performance: Deep-Dive Observations

- **Thermodynamic Dominance:** Cantera sync (48.4%) costs **20×** more than the full hydrodynamic predictor.
- **Interconnect Bound:** 63% of PCG wall time is MPI — solver is latency-sensitive to global reductions.
- **Exchange Disparity:** Energy comm (6.9%) vs species comm (0.25%) — 4-field vs 1-field halo exchange.

Optimization Priority

- 1 Batch Cantera HPY calls across cells (est. −50% energy cost).
- 2 Pipelined/CA-CG to overlap global reductions with compute.
- 3 Fuse PreFlux + Final energy exchanges into one non-blocking op.

Key Takeaways: State Conventions

Flow Density

$\rho_f = \text{params\%rho}$ always.

ρ_{thermo} is *diagnostic only*.

Energy Convention

h (sensible, relative to T_{ref}) is transported.

T is recovered via HPY inversion:

$$T = T(h, Y, p_0).$$

Architecture Warning

Do NOT couple variable-density flow by replacing ρ_f with ρ_{thermo} . The divergence constraint and pressure equation must be redesigned.

Option A Rule

After species transport, preserve h and re-solve for $T^* = T(h, Y^{n+1}, p_0)$.

Roadmap: Future Extensions

Short-term

- Batch Cantera HPY inversions.
- Pipelined PCG implementation.
- Fused energy halo exchanges.
- Regression test suite expansion.

Long-term

- Variable-density low-Mach coupling.
- Chemical reaction source terms ($\dot{\omega}_k$).
- Species-diffusion enthalpy correction.
- Radiation physics via q_{rad} .
- Higher-order bounded advection.

Questions?

`reports/technical_report/main.tex` — full technical reference